visualizations and not just static ones. You'll make a dynamic (or animated) data visualization before you finish this chapter, and using D3 to make a chart interactive is incredibly simple. A few interactive touches can make a visualization not only more readable but significantly more engaging. Users who feel like they're exploring rather than reading, even if only with a few mouseover events or a simple click to zoom, will find the content of the visualization more compelling than in a static page. But this added complexity requires an investment in learning principles of interface design and user experience. We'll get into this in more detail in chapters 9 and 12.

1.6 Your first D3 app

Throughout this chapter, you've seen various lines of code and the effect of those lines of code on the growing d3ia.html sample page you've been building. But I've avoided explaining the code in too much detail so that you could concentrate on the principles at work in D3. It's simple to build an application from scratch that uses D3 to create and modify elements. Let's put it all together and see how it works. First, let's start with a clean HTML page that doesn't have any defined styles or existing divs.

Listing 1.6 A simple webpage

```
<!doctype html>
<html>
<head>
   <script src="d3.v3.min.js" type="text/JavaScript"></script>
</head>
<body>
</body>
</html>
```

1.6.1 Hello world with divs

We can use D3 as an abstraction layer for adding traditional content to the page. Although we can write JavaScript inside our .html file or in its own .js file, let's put code in the console and see how it works. Later, we'll focus on the various commands in more detail for layouts and interfaces. We can get started with a piece of code that uses D3 to write to our web page.

Listing 1.7 Using d3.select to set style and HTML content

```
d3.select("body").append("div")
  .style("border", "1px black solid")
  .html("hello world");
```

We can adjust the element on the page and give it interactivity with the inclusion of the `.on()` function.

Listing 1.8 Using d3.select to set attributes and event listeners

```
d3.select("div")
  .style("background-color", "pink")
```